

Biodegradable Polyurethane Carrying Antifoulants for Inhibition of Marine Biofouling

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Supporting Information

ABSTRACT: Biodegradable polyurethane with *N*-(2,4,6-trichlorophenyl)maleimide (TCPM) pendant groups has been prepared via a combination of a thiol–ene click reaction and a condensation reaction. The TCPM moieties acting as antifoulants are released as the polyurethane degrades in the marine environment. The biodegradation and hydrolyzation of the polyurethane were investigated by use of quartz crystal microbalance with dissipation (QCM-D) and hydrolysis experiments. It shows both the enzymatic degradation rate and the hydrolyzation rate decrease with TCPM content, which facilitates increasing the duration of the polyurethane. Marine field tests reveal that the polyurethane has good antifouling ability since the degradation leads to a self-renewal surface and the release of the antifoulants is controlled.

1. INTRODUCTION

Biofouling is the colonization of a diverse set of microorganisms, plants, algae, and animals on submerged surfaces. It has adverse effects on marine and aquatic industries.^{1–3} Coatings containing biocides had been used to inhibit biofouling for a long time, but the global ban of harmful biocidal coatings containing tributyltin (TBT) changed the situation. Today, it is urgent to develop environmentally friendly anti-biofouling systems. For this purpose, low surface energy elastomers,^{4–6} bioinspired engineered topographies,^{7–10} zwitterionic polymers^{11,12} and electroactive polymers¹³ have been prepared. Meanwhile, some organic antifoulants from agriculture^{14,15} and natural products^{16–20} were developed.

Generally, antifoulants are physically mixed with polymer matrices to prevent biofouling growth.^{21–24} Only when their concentration is sufficient can they exhibit antifouling ability. Moreover, the release of antifoulant depends on the erosion of the polymer matrix. We have mixed the environment-friendly antifoulant (4,5-dichloro-2-*n*-octyl-4-isothiazolin-3-one) with biodegradable polyurethane. The system exhibits good antifouling ability and relatively long duration.²⁵ However, the content of added antifoulant is limited due to the immiscibility of the antifoulant and polymer matrix.

Chemical incorporation of antifoulants into a biodegradable polymer chain is expected to sustain the release of the antifoulants and improve the antifouling ability as well as the service life. Furthermore, the antifoulant loading and release rate can be tuned. It is reported that chemical incorporation of eugenol into poly(ethyl methacrylate) yields nonleaching antimicrobial materials.^{26,27} Poly(dimethylsiloxane) (PDMS) elastomers grafted with triclosan exhibits reduced fouling activity.^{28,29} Biodegradable poly(anhydride esters) containing phenolic compounds (carvacrol, thymol, or eugenol) shows antibacterial activity; however, their hydrolytic degradation is so fast that they do not have a long service life.³⁰

In the present work, we report a novel polyurethane with poly(ϵ -caprolactone) (PCL) segments in the main chain and *N*-(2,4,6-trichlorophenyl)maleimide (TCPM) or antifoulant moieties as pendant groups. PCL segments are able to degrade in marine environment due to the attack of seawater and microorganisms, leading to a controlled release of the antifoulants. We have investigated the enzymatic degradation, hydrolysis, and antifouling performance of the polyurethane. Our aim is to develop a long-acting anti-biofouling system.

2. EXPERIMENTAL SECTION

Materials. Poly(ϵ -caprolactone)diol (PCL, M_w = 2000 g/mol) was from Perstorp and dried under reduced pressure for 2 h prior to use. *N*-(2,4,6-Trichlorophenyl) maleimide (TCPM) from Linsheng Chemical Co. was used as received. 3-Mercapto-1,2-propanediol (TG) and 2,2-dimethoxy-2-phenylacetophenone (DMPA) were from Aladdin and used as received. 1,4-Butane diol (1,4-BD) from Aldrich was dried under reduced pressure for 2 h prior to use. Dibutyltin dilaurate (DBTDL) from Aldrich was used as received. L-Lysine ethyl ester diisocyanate (LDI) from Dahong Chemical was used as received. Tetrahydrofuran (THF) was refluxed over CaH₂ and distilled prior to use. Lipase PS purchased from Aldrich was used as received. Artificial seawater was prepared following ASTM D1141-98 (2003). Other reagents were used as received. The synthesis of biodegradable polyurethane with TCPM pendant groups is illustrated in Scheme 1.

Synthesis of TCPM(OH)₂. TCPM(OH)₂ was prepared via a thiol–ene click reaction.³¹ Typically, 1.73 g of TG (16 mmol), 1.11 g of TCPM (4 mmol) and 0.02 g of DMPA (0.08

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high temperature (340–390 °C) in comparison with PU-N0 (Figure S5 in the SI). Moreover, the decomposition temperature ($T_{50\%}$) increases with TCPM content (Table 1). Accordingly, the incorporation of antifoulant moieties improves the thermal stability of the polyurethane.

Enzymatic Degradation. Enzymatic degradation was monitored by quartz crystal microbalance with dissipation (QCM-D) and surface plasmon resonance (SPR). PU-Nx films were prepared by spin-casting of PU-Nx solution in THF solution (5.0 mg/mL) on a spin-coater (CHEMAT, KW-4) at 4000 rpm in air. QCM-D and the AT-cut quartz crystal with a fundamental resonant frequency of 5 MHz were from Q-sense AB (Sweden). The quartz crystal was mounted in a fluid cell with one side exposed to the solution. The effects of surface roughness were minimized by using highly polished crystals with a root-mean-square roughness less than 3 nm. The details about QCM-D measurements can be found elsewhere.³² Briefly, QCM-D simultaneously monitors changes in resonance frequency (Δf) and dissipation (ΔD) in real time. Δf is related to the change in the mass attached to the oscillating sensor surface, whereas ΔD is related to the change in the viscoelasticity of the adsorbed layer. For a rigid film in vacuum or air, if it is evenly distributed and much thinner than the crystal, Δf is related to Δm and the overtone number ($n = 1, 3, 5 \dots$) by the Sauerbrey equation,³³

$$\Delta m = -\frac{\rho_q l_q \Delta f}{f_0 n} = -C \frac{\Delta f}{n} \quad (1)$$

where f_0 is the fundamental frequency, ρ_q and l_q are the specific density and thickness of the quartz crystal, respectively, and C is the constant of the crystal (17.7 ng/cm²·Hz). In the present study, artificial seawater was used as the reference, and the Lipase PS solution (0.5 mg/mL) was delivered to the surface at a flow rate of 150 μ L/min. The changes in frequency (Δf) and dissipation (ΔD) gave the information about the degradation and structural change of the PU-Nx films. All the experiments were performed at 25 °C, and the presented data were from the third overtone ($n = 3$). The change of mass on the resonator surface was estimated by using QTools software (Q-Sense AB) in terms of Voigt viscoelastic model.³²

SPR measurements were performed on SPR Navi 210A (Bionavis) instrument equipped with an autosampler accessory at 25 °C. The amount of mass change on the gold sensor slide can be followed by monitoring the change in angular position (angular scan mode) over time.³⁴ The SPR angle shift is linear to the added mass of the layer with $0.1^\circ \approx 1$ ng/mm². The Lipase PS solution (0.5 mg/mL) was delivered to the surface at a flow rate 10 μ L/min.

Hydrolytic Degradation. The PU-Nx film was prepared via a solution casting method. Typically, a solution of 30% (w/v) polyurethane in THF was dripped onto a glass fiber reinforced epoxy resin panel (20 mm $\times$ 20 mm in size). The panels coated with PU-Nx were kept at room temperature for 3 days to remove the residual THF. Then, the weight (W_0) of each dried coating together with its panel was measured before dipping into a tank of artificial seawater that was renewed every 2 weeks. After a certain period of time, the panel was taken out, rinsed with deionized water three times, and dried at 60 °C in a vacuum oven, and the weight (W_1) of the panel was measured again. The mass loss was designated as $(W_1 - W_0)/\text{test area}$. For each sample, three coated panels were prepared and

measured, and each data point was averaged over three successive and consistent measurements.

Marine Field Tests. The field tests were performed starting from May 7, 2013 at the inner Xiamen bay (24°45'N, 118°07'E) in China, where the marine biofouling was heavy because of the rich fouling organisms. The samples applied onto the glass fiber reinforced epoxy resin panels (250 $\times$ 100 $\times$ 3 mm³) were lowered into seawater at depths of 0.2 to 2.0 m. After a certain period of time, the panels were taken out of the sea and carefully washed with seawater and photographed, and then they were immediately placed into the seawater to continue the test. Panels coated with PU-Nx were tested. The antifouling activity was evaluated according to ASTM D 6990-05 (2011).³⁵ The range of the fouling rating (FR) was 0 to 100. The FR for a coating free of biofouling settlement was recorded as 100. The FR of each coating was obtained by deducting from 100, depending on the percentage of area covered by macrofouling, that is, the fouling rating essentially reflects the nonfouled area. In the current experiment, the fouling community mainly includes the barnacles, algae, and slime. Distances smaller than 1 cm from the edge of the panels were not considered.

3. RESULTS AND DISCUSSION

Figure 1 shows the DSC curves of PU-Nx. For each PU-Nx sample, we can observe only one glass transition, and the T_g

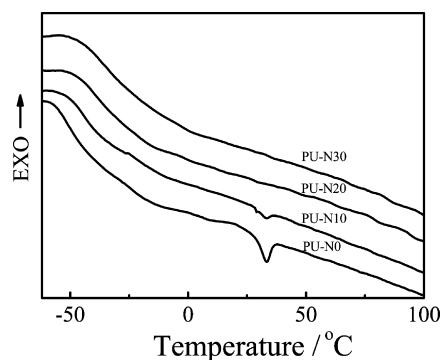


Figure 1. DSC curves of PU-Nx.

increases with antifoulant content. The facts indicate that antifoulant moieties are randomly distributed along the polyurethane chain. On the other hand, a melting peak around 34 °C can be observed in PU-N0 due to the semicrystalline PCL segments.³⁶ The incorporation of antifoulant leads the melting peak to be smaller. When the antifoulant content is above 20 wt %, the peak disappears. Clearly, the incorporation of antifoulant destroys the crystalline structure of PU-Nx, which would increase the degradation rate.

Considering that a large population of microorganisms such as bacteria, actinomycetes, and fungi in natural marine environments can produce enzymes,^{37,38} we first examined the enzymatic degradation by using of QCM-D. Figure 2 shows the time dependence of frequency shift (Δf) for the enzymatic degradation of the PU-Nx in artificial seawater. For PU-N0, after lipase PS is introduced, Δf increases and gradually levels off. After rinsing with seawater, Δf exhibits a marked increase relative to the baseline. It is known that the increase of mass on the sensor surface causes the frequency to decrease.³⁹ The increase in Δf indicates the mass loss of the polyurethane film. Namely, the PU-Nx films degrade into small molecules and

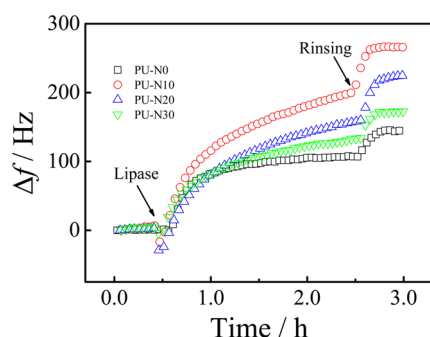


Figure 2. Time dependence of the frequency shift (Δf) for the enzymatic degradation of PU-N x in artificial seawater at 25 °C.

disperse in artificial seawater. For PU-N10, a rapid increase in Δf is observed, indicating a faster degradation. This is because the incorporation of antifoulant moieties reduces the crystallinity of the polyurethane. Further increasing the antifoulant content leads Δf to be smaller (for PU-N20 and PU-N30), indicating that the degradation becomes slower. This is understandable because PCL moieties decrease as the antifoulant content increases (Figure S3 in the SI). We also examined PU-N x with the same PCL content. Its degradation increases with the antifoulant content (not shown). Thus, the enzymatic degradation rate of PU-N x can be tuned by the antifoulant content.

Figure 3 shows the time dependence of the energy dissipation shift (ΔD) for the enzymatic degradation of the

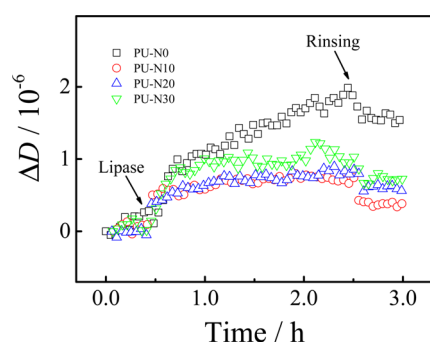


Figure 3. Time dependence of the energy dissipation shift (ΔD) for the enzymatic degradation of PU-N x in artificial seawater at 25 °C.

PU-N x films in artificial seawater. It is known that the dissipation of a polymer layer on the quartz resonator increases with its thickness and looseness. Namely, a dense and rigid structure leads to a small dissipation of energy, whereas a looser and thicker structure results in a larger dissipation.⁴⁰ For all the PU-N x films, ΔD increases with time. PU-N0 has the largest ΔD value. As we know, PU-N0 has the highest crystallinity. The enzyme first attacks the amorphous region and then the crystalline area. Since the former is faster than the latter, a random porous structure results with a large ΔD . For PU-N10, the difference of degradation rates between amorphous and crystalline regions becomes smaller as the crystallinity decreases, so that we observe a lower ΔD compared with that of PU-N0. However, further increasing the antifoulant content (PU-N20 or PU-N30) leads the enzymatic degradation to decrease, leaving a thicker film, so that ΔD slightly increases with antifoulant content.

The enzymatic degradation was also investigated by SPR (Figure 4). After lipase PS solution is introduced, the SPR angle

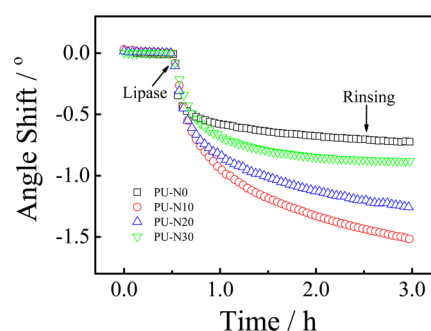


Figure 4. Time dependence of SPR angle shift for the enzymatic degradation of PU-N x in artificial seawater at 25 °C.

sharply decreases and then gradually levels off, indicating the mass loss of the film. For PU-N0, the decrease in SPR angle after rinsing is about 0.73°. As the antifoulant content increases to 10 wt % (PU-N10), a larger angle shift can be observed because PU-N10 with a lower crystallinity degrades faster than PU-N0. However, further increasing the antifoulant content (for PU-N20 and PU-N30) leads the angle shift to decrease in that the degradation decreases due to the decrease of PCL content. All these are consistent with the QCM-D results.

The mass loss of the PU-N x samples can be estimated by SPR shift. For PU-N0, PU-N10, PU-N20, and PU-N30 in 3 h, the mass loss values are 7, 15, 13, and 9 ng/mm², respectively. They can also be estimated from QCM-D measurements in terms of the Voigt model. For PU-N0, PU-N10, PU-N20, and PU-N30, they are 11, 21, 17, and 12 ng/mm², respectively. Clearly, the mass loss obtained by SPR is less than that from QCM-D. This is understandable because the latter represents hydrodynamic mass that includes those of the polymer and the coupled water,³² whereas the former is the mass of the polymer only.

Figure 5 shows the time dependence of mass loss of PU-N x in artificial seawater at 25 °C. After immersing in artificial

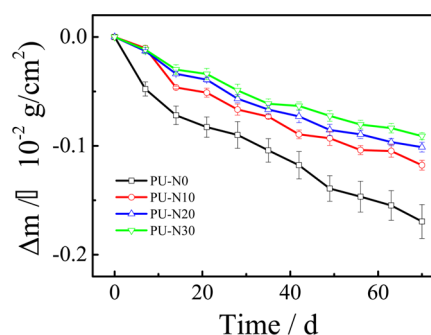


Figure 5. Time dependence of mass loss of PU-N x in artificial seawater at 25 °C.

seawater for 7 days, all PU-N x samples start to lose weight. Generally, the mass loss is almost linearly related to time except for that at the early stage, indicating the degradation is uniform. In comparison with PU-N0, the PU-N x samples with antifoulant generally lose their weight more slowly, indicating a slower hydrolytic degradation. Besides, the mass loss decreases with the antifoulant content (PU-N10 to PU-N30) since the PCL content in PU-N x decreases with the increasing

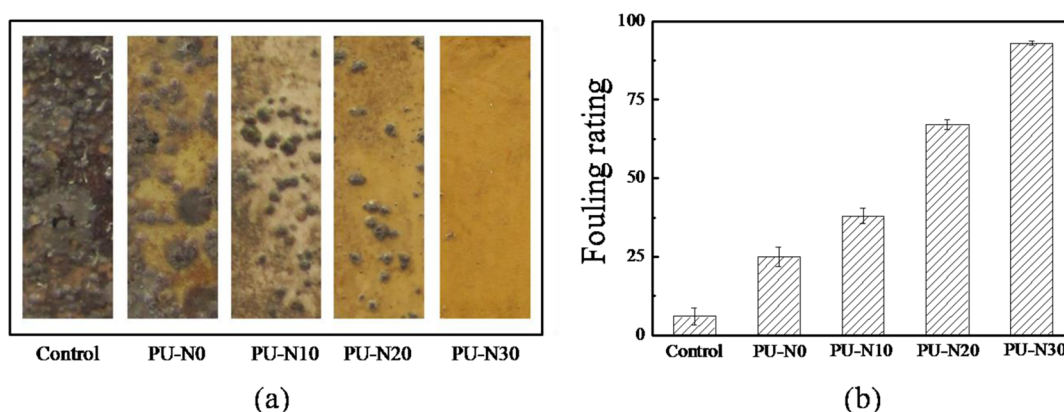
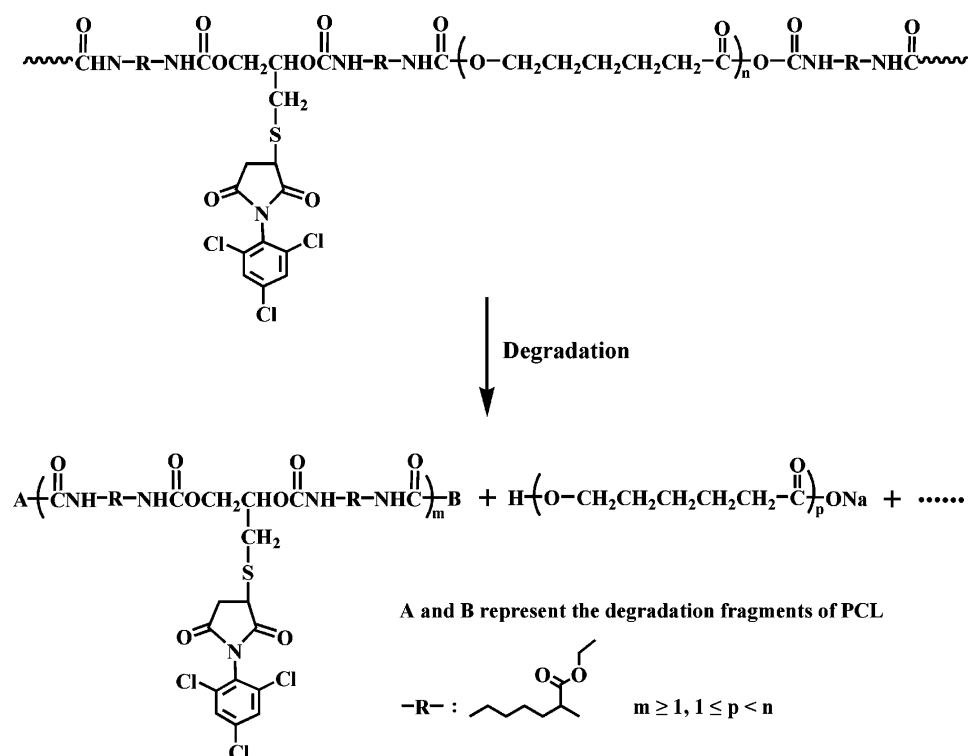


Figure 6. (a) Typical images of panels coated with PU-N x after immersion in seawater (May–July 2013, Xiamen, China) for two months. (b) The fouling rating calculated according to ASTM D 6990-05 (2011). Each error bar represents the standard deviation calculated from three consecutive measurements.

Scheme 2. Possible Hydrolysis Reaction of the Biodegradable Polyurethane with Antifoulant Pendant Groups



antifoulant content (Figure S3 in the SI). Moreover, the hydrophobic phenyl groups of antifoulant in PU-N x decrease the interaction between PU-N x and water. As a result, the polyurethane with higher antifoulant content exhibits a lower mass loss. Nevertheless, biodegradable polyurethane with antifoulant pendant groups sustains the release of antifoulant even when the antifoulant content is as high as 30 wt %.

The anti-biofouling of PU-N x was examined by marine field testing. Figure 6 shows the typical images of panels coated with PU-N x (a) and the corresponding fouling rating after immersion in seawater for two months (b). The control panels with epoxy resin surfaces are seriously fouled after two months. For the panels coated with PU-N0, less fouling on the surface can be observed. Namely, it can inhibit the biofouling to some degree. As reported before, the surfaces constructed with a biodegradable polymer are gradually decomposed and eroded

under the attack of seawater or enzymes in marine environments, leading to a self-renewal surface that can polish the attached living organisms or inorganics if the polishing is faster than their attachment and growth.^{25,41} That is probably why PU-N0 has some anti-biofouling ability. Obviously, the anti-biofouling only from degradation is limited. For the panels coated with PU-N x having antifoulant moieties, the fouling rating dramatically increases as the antifoulant content increases, indicating the enhanced antifouling performance. Note that both the enzymatic degradation rate and the hydrolyzation rate decrease with the antifoulant content. Actually, the amount of the released antifoulant depends on not only the degradation rate but also the antifoulant content in the polyurethane. Thus, the enhanced antifouling ability arises from the increased antifoulant. In other words, the polyurethane with higher antifoulant content can release enough

antifoulants even though its degradation rate decreases, which inhibits the location and growth of marine microorganisms. Another possible reason is that the nonleaching TCPM on the PU-Nx surface also has antifouling ability.^{28,29} Accordingly, as the amount of antifoulant in the coating increases, the antifouling activity increases though the degradation rate of the coating decreases. Moreover, the polyurethane with a slow degradation has a long service life. Thus, the anti-biofouling performance and the duration of the polyurethane can be controlled by the composition. Considering that the degradation products are complex, it is hard to quantify the released antifoulant so far. We will manage to do that in the future.

In marine environments, there exist some inorganics besides marine organisms. They can form a thin film on a surface for biological species to accommodate.¹ A self-renewal surface can prevent either marine organisms or inorganics from locating and growing. The faster change of the surface leads to a better anti-biofouling ability but shorter service life. Here, the antifoulants are chemically incorporated into a biodegradable polyurethane. The degradation of the polyurethane makes the self-renewing of the surface and controlled release of the antifoulants possible. Actually, the hydrolysis reaction of PU-Nx coating includes the cleavage of the ester bond in PCL segments and the urethane group (NHCOO). The latter is relatively stable, and its cleavage happens under conditions such as strong alkali solution or fungus.^{42,43} We detected the degradation products by using liquid chromatography–mass spectrometry (LC–MS). We found the fragments of PCL in the degraded products. Thus, the degradation occurs at the ester groups in PCL segments (Scheme 2). Accordingly, the biodegradable polyurethane with antifoulant pendant groups is expected to have not only high anti-biofouling performance but also long service life in marine environments. Moreover, the current PU-Nx coating is free from heavy metals so that it will not cause electrochemical corrosion and heavy metal pollution. TCPM with low toxicity ($Ld_{50} > 4500$ mg/kg) has been used as an environment-friendly marine antifouling agent.^{24,44} The PU-Nx with PCL segments can degrade into oligomers and finally decompose into carbon dioxide and water in the environment.^{37,42} Thus, the coating is not harmful to our environment.

4. CONCLUSION

We have prepared polyurethane with poly(ϵ -caprolactone) (PCL) segments in the main chain and antifoulant moieties as pendant groups. The polyurethane can degrade in the marine environment, and the degradation rate can be tuned by its composition. The polyurethane shows anti-biofouling performance depending on the release of the antifoulants. The anti-biofouling ability and duration can be controlled by the composition of the polyurethane. The present study demonstrates that biodegradable polymers with chemically incorporated antifoulants are expected to find applications in marine anti-biofouling.

■ ASSOCIATED CONTENT

Supporting Information

HRMS for TCPM(OH)₂, ¹H NMR, FTIR spectra of PU-Nx, and TGA curves of PU-Nx. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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